

SVM Decision Boundary Code Explanation

1. Import Libraries:

NumPy is used for numerical operations.

Matplotlib is used for plotting graphs.

SVC is the Support Vector Classifier model.

2. Feature Selection:

We select 'income' and 'credit_score' as input features.

Target variable is 'approved'.

3. Model Training:

SVC with linear kernel is used to create a straight decision boundary.

4. Scatter Plot:

Data points are plotted using income (X-axis) and credit score (Y-axis).

Colors represent approved/rejected classes.

5. Grid Creation:

We create a grid of points using linspace and meshgrid to cover the entire graph.

6. Decision Function:

The model predicts values for each grid point.

Positive values belong to one class, negative to another.

7. Contour Plot:

Contour is drawn at level 0 which represents the decision boundary.

8. Final Output:

Graph shows separation between approved and rejected loan customers.